

STILLSTACK · REQUIREMENTS

StillStack requirements

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DOCUMENT CONTROL

Document ID	SSK-REQ-001	Version	0.2
Status	Draft	Date	2026-09-24
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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

StillStack requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Produce more distillate per unit area than a basin still	10 L per m ² of aperture per day or more on a clear day with 5.5 kWh per m ² of sun (design goal 13 L)	Stage energy balance calculation; later outdoor test with a single-basin still alongside
R2	Reuse the latent heat of condensation	Daily gained output ratio (latent heat in distillate divided by solar input) of 1.3 or more	Stage energy balance calculation
R3	Remove salt	Distillate conductivity 75 μ S/cm or less (about 50 mg/L dissolved solids) from feed up to 40 g/L salinity	Calculation of carryover paths; later conductivity meter
R4	Keep brine and distillate apart	A physical air break of 10 mm or more between every wick edge and every distillate channel; no shared drain	Design review of the model
R5	Run without power	No pumps, electronics or batteries; gravity feed from a user-filled container at most 1.5 m above ground	Design review
R6	Accept real feed water	Seawater or brackish water up to 40 g/L salinity after cloth or 20 μ m prefiltering; feed-to-distillate ratio 2.5 or more so salt stays in solution	Salt balance calculation
R7	Food-safe wetted parts	Every surface that touches distillate is food-contact rated (for example 316 stainless steel, food-grade coated aluminum, PP, HDPE or NSF/ANSI 51 or 61 silicone)	Material list review
R8	Survive dry stagnation	No damage after an empty-wick day at 1,000 W/m ² and 35 °C ambient; all internal materials rated to 110 °C or more	Stagnation temperature calculation; material datasheets
R9	Household scale and portable	1.0 m ² aperture; panel mass 20 kg or less dry; carried by two people; footprint within 1.3 x 1.3 m	Massing model and mass estimate
R10	Easy maintenance	Wicks removable, rinsed and refitted by one person without special tools in 30 min or less; salt flushing no more often than once a week	Design review
R11	Adjustable tilt	Tilt between 10 and 35 deg to suit latitudes of about 0 to 35 deg	Model check
R12	Low cost and buildable	Parts cost \$250 or less for the 1 m ² prototype; hand tools, sheet metal shears and a 3D printer only	Priced BOM

■ Assumptions

- Design day: 5.5 kWh per m² per day (19.8 MJ per m²) on the tilted aperture, 30 °C mean daytime ambient, light wind.

- Latent heat of vaporization taken as 2.26 MJ per kg for first-order estimates (the true value at 40 to 70 °C is about 2.33 to 2.41 MJ per kg, which makes the estimates slightly optimistic).
- The WHO does not set a health-based limit for dissolved solids; 50 mg/L is chosen as a check that salt carryover is negligible, not as a potability limit.
- Household need of about 3 L per person per day for drinking and 2 L for cooking, so 13 L per day serves a household of two or three.